Embedded Wireless Motor Monitoring and Control System

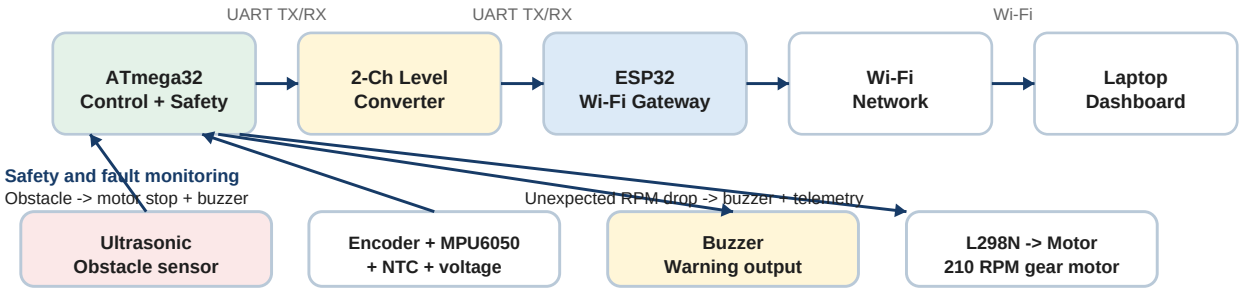
Graduation Project - Revised Materials, Functions, and System Architecture

Project concept

A low-cost embedded system built around an **ATmega32** motor controller and an **ESP32** wireless gateway. The ATmega32 performs real-time motor control, encoder feedback, sensor acquisition, safety handling, and fault detection. The ESP32 provides Wi-Fi connectivity to a laptop/dashboard. The controller link is now a direct **UART TX/RX connection through a 2-channel bi-directional 3.3/5 V logic-level converter**, replacing the earlier RS-485 architecture.

System architecture

ATmega32 -> 2-Channel Bi-directional Logic Level Converter -> ESP32 -> Wi-Fi -> Laptop / Dashboard



Motor path: ATmega32 -> L298N -> 25GA370 12 V **210 RPM** DC gear motor with encoder. Feedback path: encoder + MPU6050 + NTC thermistors + voltage sensing -> ATmega32. Safety path: ultrasonic sensor -> ATmega32 -> motor stop + buzzer warning.

1. Core controller and motor hardware

Component	Qty.	Importance	Function
ATmega32 development board	1	Critical	Main real-time controller. Handles PWM, encoder counting, ADC, timers, interrupts, I2C/SPI/UART, safety logic, and the motor-control algorithm.
ESP32 development board	1	Critical	Wireless/network gateway. Receives data from the ATmega32 and sends it over Wi-Fi to the laptop/dashboard.
25GA370 12 V 210 RPM DC gear motor with encoder	1	Critical	Provides the physical motor load. The encoder supplies rotational feedback so RPM can be measured and motor-speed control/fault detection can be implemented.
L298N motor driver module	1	Critical	Power interface between the ATmega32 and motor. Provides direction control and PWM speed control.

2. Sensors, feedback, and monitoring

Component	Qty.	Importance	Function
MPU6050 accelerometer/gyroscope module	1	High	Measures acceleration and angular motion for vibration/condition monitoring of the motor assembly.
NTC 10K thermistor	2	High	Temperature sensing. One can monitor motor temperature and the other the electronics/driver area.
Resistor voltage-divider set	1 set	Medium	Scales a higher supply voltage to a safe ATmega32 ADC range when direct voltage monitoring is required.

Component	Qty.	Importance	Function
Voltage sensor module (as listed in component sheet)	1	High	Voltage-only monitoring option retained after removal of the INA219 current/voltage sensor. Interface to the ATmega32 ADC according to the module specification.
Ultrasonic distance sensor	1	Critical	Safety sensor. When an object enters the configured safety distance, the ATmega32 stops the motor and activates the buzzer warning.
Motor encoder (integrated)	1	Critical	Provides pulse feedback for measured RPM. Firmware compares measured RPM with the commanded/reference speed to detect an unexpected speed drop.

3. UART communication between ATmega32 and ESP32

Component	Qty.	Importance	Function
2-Channel Bi-directional Logic Level Converter (3 V - 5 V)	1	Critical	Provides two bidirectional level-shifted channels for UART TX/RX. ATmega32-side logic is 5 V; ESP32-side logic is 3.3 V.
UART TX/RX connection set	1	High	Carries serial data between the ATmega32 and ESP32. TX and RX are crossed between the two controllers through the level converter.

Electrical note: Do not connect a 5 V ATmega32 UART signal directly to an ESP32 GPIO. The two UART lines should pass through the 2-channel bi-directional logic-level converter.

4. Power and filtering

Component	Qty.	Importance	Function
12 V DC power supply, 2-3 A	1	Critical	Main energy source for the motor and electronics. Provides margin for startup and load changes.
LM2596 buck converter	1	Critical	Steps 12 V down to 5 V for the ATmega32-side electronics when required.
3.3 V regulator	1	Critical	Provides a clean 3.3 V rail for the ESP32 and other 3.3 V devices when an external 3.3 V supply is required.
Fuse + holder	1	High	Protects the power system from overcurrent and wiring faults.
470-1000 uF electrolytic capacitor	1-2	High	Buffers supply dips and motor-generated disturbances.
100 uF electrolytic capacitors	Several	Medium	Bulk filtering near power entry and converter outputs.
10 uF capacitors	Several	Medium	Local power stabilization and regulator input/output filtering.
100 nF ceramic capacitors	Several	Critical	High-frequency decoupling for the ATmega32, sensors, interface IC, and other digital circuits.

5. User interface and warnings

Component	Qty.	Importance	Function
Low-current active buzzer / piezo buzzer	1	Critical	Audible warning for safety and fault conditions: (1) obstacle detected by the ultrasonic sensor; (2) motor speed falls below the configured acceptable value while a higher speed was commanded.
Push buttons	3-4	Medium	Start/stop, menu selection, and parameter adjustment.
LEDs	4-6	Medium	Power, communication, normal, warning, and fault indicators.

6. Passive components and signal conditioning

Component	Qty.	Importance	Function / use
220/330 ohm resistors	Several	Medium	LED current limiting.
1K ohm resistors	Several	Medium	General signal conditioning and pull-up/pull-down or series-resistor applications as required by the final circuit.
4.7K ohm resistors	Several	High	Typical I2C pull-up resistors and other pull-up/pull-down uses.
10K ohm resistors	12 per component sheet	Critical	NTC divider circuits, pull-ups, and voltage-divider networks.
47K / 100K ohm resistors	Several	Medium	Voltage-divider and analog scaling networks.

Note: All standalone diode parts from the original list have been removed by request. This includes the TVS diode, reverse-polarity diode/protection entry, and small-signal / Schottky diodes. Small MOSFETs / transistors (including the 2N7000 items listed in the component sheet) have also been removed.

7. Prototyping, wiring, and final construction

Component	Qty.	Importance	Function
Large breadboard	1-2	High	Fast prototyping and initial circuit validation before PCB design.
Jumper wire kit	1	High	Temporary interconnections during prototyping.
Perfboard	1	Medium	Intermediate build after breadboard validation and before the final PCB.
Screw terminal blocks	Several	High	Reliable detachable connections for motor, supply, and external wiring.
Pin headers / JST connectors	Several	Medium	Connect sensors, programming, and modular boards.
Enclosure	1	Medium	Protects the final assembled electronics and gives the project a professional finish.

8. Programming and test equipment

Component	Qty.	Importance	Function
USBasp programmer	1	High	Programs the ATmega32 when an onboard programmer is not available.
USB-to-TTL serial adapter	1	High	UART debugging and PC communication during firmware development.
Digital multimeter	1	Critical	Checking voltage, continuity, resistance, current, and troubleshooting.
Oscilloscope	1 / if available	High	Checking PWM, encoder pulses, UART TX/RX, timing, and power-rail noise.

9. Safety and fault-detection behavior

Condition	ATmega32 response	Buzzer	Motor
Ultrasonic sensor detects an object inside the configured safety distance	Enter safety-stop state and flag the condition	ON / warning pattern	STOP
Measured RPM falls below the configured acceptable value while a higher speed is commanded	Flag speed-loss fault and report actual + commanded RPM	ON / warning pattern	Remain under controller logic; stop behavior can be configured separately
Normal operation	Continue closed-loop speed monitoring and sensor acquisition	OFF	RUN at commanded speed

Unexpected speed-loss logic

The motor encoder provides the measured RPM. The ATmega32 compares the measured speed with the commanded/reference speed. If the measured RPM stays below a configurable acceptable percentage or absolute threshold for a configurable time, the controller activates the buzzer and reports the fault to the ESP32. This detects external loading or another disturbance that causes the motor to slow down without a commanded reduction in speed.

Recommended firmware structure

Task	ATmega32 role
Motor control	Generate PWM and direction commands for the L298N.
RPM measurement	Count encoder pulses using timer/interrupt resources and calculate actual RPM.
Speed-fault check	Compare actual RPM against commanded/reference RPM using configurable limits and delay.
Obstacle safety	Read ultrasonic distance and enter a motor-stop state when the configured safety distance is crossed.
Warning output	Drive the buzzer for obstacle and unexpected speed-loss events.
Sensor acquisition	Read MPU6050, NTC thermistors, and voltage sensing.
UART telemetry	Send RPM, temperature, sensor state, safety/fault state, and commanded values to the ESP32.
ESP32 gateway	Receive ATmega32 data, provide Wi-Fi, and expose the data to a laptop/dashboard.

10. Final consolidated material list

Category	Component	Qty.
Control	ATmega32 development board	1
Control	ESP32 development board	1
Motor	25GA370 12 V 210 RPM DC gear motor with encoder	1
Motor	L298N motor driver module	1
Sensors	MPU6050 module	1
Sensors	NTC 10K thermistor	2
Sensors	Ultrasonic distance sensor	1
Sensors	Voltage sensor module (component sheet)	1
Sensors	Resistor voltage-divider set	1 set
Communication	2-channel bi-directional 3.3/5 V level converter	1
Power	12 V DC power supply, 2-3 A	1
Power	LM2596 buck converter	1
Power	3.3 V regulator	1

Category	Component	Qty.
Protection	Fuse + holder	1
Filter	470-1000 uF electrolytic capacitors	1-2
Filter	100 uF electrolytic capacitors	Several
Filter	10 uF capacitors	Several
Filter	100 nF ceramic capacitors	Several
Interface	Low-current active buzzer / piezo buzzer	1
Interface	Push buttons	3-4
Interface	LEDs	4-6
Resistors	220/330 ohm resistors	Several
Resistors	1K ohm resistors	Several
Resistors	4.7K ohm resistors	Several
Resistors	10K ohm resistors	12 listed
Resistors	47K/100K ohm resistors	Several
Build	Breadboard / jumper wires / perfboard	As required
Build	Screw terminals / headers / enclosure	As required
Test	USBasp / USB-to-TTL / multimeter / oscilloscope	As available

11. Items removed or replaced from the previous version

Previous item	Action	Reason / new arrangement
INA219 current/voltage sensor	REMOVED	Removed as requested. Voltage monitoring is handled separately through the retained voltage-sensing path.
MAX485 / RS-485 transceivers	REMOVED	Replaced by direct UART TX/RX between ATmega32 and ESP32 through a 2-channel 3.3/5 V logic-level converter.
Twisted-pair RS-485 cable	REMOVED	No RS-485 bus in the revised architecture.
120 ohm RS-485 termination resistors	REMOVED	No RS-485 bus in the revised architecture.
16x2 LCD with I2C backpack	REMOVED	Laptop dashboard over Wi-Fi replaces the local LCD.
10K potentiometer	REMOVED	No manual potentiometer speed reference is required in the revised design.
TVS diode	REMOVED	All standalone diode parts removed by request.
Reverse-polarity protection entry	REMOVED	All standalone diode/protection-diode entries removed by request.
Small-signal / Schottky diodes	REMOVED	Removed by request.
Small MOSFETs / transistors	REMOVED	Removed by request; the 2N7000 items from the component sheet are therefore not included.
25GA370 12 V 130 RPM motor	UPDATED	Same motor family, updated to the requested 210 RPM version with encoder.

12. Component-sheet items incorporated

The provided component sheet was used to update the consolidated list. The 10K resistor quantity is carried as **12 pieces**; the 25GA370 motor entry is updated from the listed 130 RPM version to the requested 210 RPM version; the L298N, MPU6050, NTC thermistor, capacitors, and other surviving passive items are retained. The listed 2N7000 and TVS-diode entries are intentionally excluded because you requested removal of MOSFETs/transistors and all diode parts.

Key component-sheet items incorporated: 12 x 10K ohm resistors; the 25GA370 motor family (updated from the listed 130 RPM version to the requested 210 RPM version); L298N; MPU6050; NTC 10K thermistors; and the listed capacitor parts. The 2N7000 and TVS-diode entries are intentionally excluded because of the requested removals.